

The IoT helps in the inter-networking of different physical structures and hardware devices—buildings, vehicles, electronic gadgets and other devices that are embedded with the actuators, sensors and software, so that they can collect and exchange data between each other. While the Internet of Things is still in its infancy, it has become very clear that it will soon be a part of everyone's life, if that is not already the case. As different companies realise the revolutionary potential of the IoT, they have started finding a number of obstacles they need to address to leverage it efficiently. Many businesses and industries use machine learning and, more specifically, the ML-as-a-Service (MLaaS) to exploit the IoT's potential.

Machine learning is basically a part of computer science that makes any system smart enough to learn on its own without actually being programmed for that task. It helps a system or device learn in the same way as humans learn by themselves. As we learn any type of system on the basis of our experience and the knowledge that is gained after analysing it, machines too can analyse and study the behaviour of a system or its output data and then learn how to take different decisions on that basis. Recently, there have been many factors that have come together to make ML a reality — large data sources, the increased computational power required for processing the information in split seconds, and different algorithms that have become more reliable. Machine learning can be used in different scenarios such as when the required outcome is known (supervised learning), when data is not known beforehand (unsupervised learning), or when learning depends upon the result of the interaction between any specific model and the environment (reinforcement learning).

The main purpose behind ML is to automate the development of different analytical models to enable algorithms to continuously learn with the help of available data. Google's self-driving vehicle is one such development that uses different ML techniques with IoT to create a completely autonomous vehicle. It combines the advanced features of different modern cars (like speech recognition, lane assistance, adaptive cruise control, parking assistants and navigators).

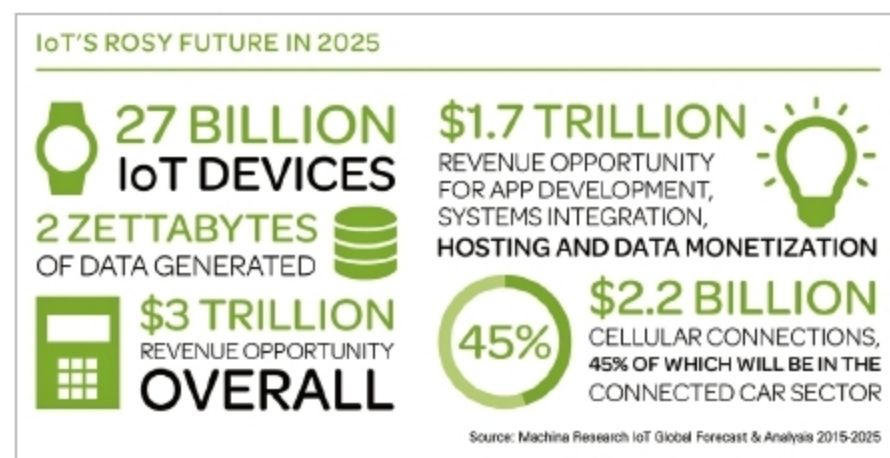


Figure 1: The future of IoT (Image source: *googleimages.com*)w

IoT without ML

IoT undoubtedly uses the Internet in an amazing manner, but there are still some challenges, particularly if it's implemented without ML. Humans have never played around with so many devices generating such huge amounts of diversified data, ever before. Let's have a look at some of the challenges facing IoT without ML.

Device diversity and interoperability: There are so many industries and companies offering products and services in a single domain. In the case of a smart grid, there are so many kinds of sensors that measure the power consumption from various organisations and corporations, all operating based on different standards. To keep all such devices and sensors working together is really a big challenge, which ML can potentially overcome.

Device management: We know that there are a large number of small devices connected to any smart system. All of them need to communicate with each other and also to servers spread over large geographical areas. So in all such cases, there are possibilities that devices may not be able to connect with each other—there might be several data linking issues that need to be managed efficiently. For instance, you may want to open a front door remotely but the command used to open the door needs to be transmitted through the light in front of the door. This requires all devices to be managed in an elastic topology such that the communication between each of them happens smoothly.

Integration of data from multiple sources: When we deploy an IoT application, we will inevitably gather abundant diversified data from different sources such as sensors and actuators, as well as contextual data from various mobile devices, social network feeds and so on. To create and build the connection between all that data can really be a big challenge without ML.

Scalability, data volumes and performance: We need to prepare our businesses to manage the scalability, the large data volumes, as well as the speed at which different devices connected to the IoT generate the data. This is actually a common Big Data problem to be dealt with. For IoT, the analysis and handling of data needs to be in near real-time.

Flexibility and how applications evolve: Different devices and sensors will continue to evolve with new capabilities and improved functions. This will lead to the creation of new use cases and also new business models. We will need to quickly develop our products with minimal effort using certain frameworks in order to catch up with the pace of changing techniques. To enable and sustain this, we need different ecosystems and platforms to support us. Machine learning can actually provide this.

How ML increases the efficiency of IoT

1. There are some data models that implement traditional data analytics, and these are often static and of limited use when it comes to addressing rapidly changing and